

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 87

THE QUINN HYDRONIC ACTUATOR GEOTHERMAL CORES & ARCHIMEDES TWIN-SCREW PUSH-PULL PUMPS

Fluid muscle actuators run silent on geothermal cores
twin screws push and pull on one drive line



the muscle is water — the pump is two screws

HYDRONIC MUSCLE · $P = F/A$ · PUSH-PULL SCREWS

PUSH AND PULL — ONE DRIVE LINE

Geothermal cores — v1.0 in force

GP-IM-087

Revision v1.0 — 22 August 2026

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VETTED — GEMINI + KIMI + CHATGPT — 22 AUGUST 2026 — SHIP IT

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 87

PHASE 87: THE QUINN HYDRONIC ACTUATOR GEOTHERMAL CORES & ARCHIMEDES TWIN-SCREW PUSH-PULL PUMPS — STATUS: CURRENT. Cold grease is a broken arm waiting for a load. Phase 87 rejects local exterior joint-heating jackets and centralizes the problem: a hyper-insulated structural Lub-Sump Tank inside the primary vehicle chassis, its high-efficiency thermal elements pre-heating heavy-duty industrial greases to low-viscosity working fluid before any arm line is asked to move. The circulation is mechanical, not hydraulic-pressure-dependent: dual Archimedes screw feeds embedded in the core of the arm lines — Screw A pushes the heated fluid grease outward, Screw B pulls the return — overriding the pump blowouts that sub-zero sludge-viscosity causes in conventional hydraulic circuits. And the system audits itself: a widened bulbous diagnostic terminal on the return conduit houses inline pressure sensors and micro-kinetic displacement testers measuring real-time flow and viscosity baselines, with motor ignition interlocked to the audit — the arm cannot be driven cold. The vet graded it **PASS**: hydraulic actuators are ordinary fluid power, Archimedes screws are ordinary positive-displacement machines — the questions are pressure, flow, seals and temperature, and they are owed in §9.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-087, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the hydronic law as seated (contents line, the three design bodies — verbatim) — §2 why it works (fluid power and positive displacement — graded) — §3 the system in the arm lines — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 87: **Hydronic Actuator Geothermal Cores & Archimedes Twin-**

Screw Push-Pull Pumps. Engineers high-pressure joint actuators running on sleeved and heated fluid loops. Utilizes heavy-duty, axle-less Archimedes twin-screw push-pull pumps to sustain continuous multi-directional hydraulic pressure under sub-zero operations. Stops arctic region and space equipment joints freezing'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Sub-Zero Actuator Joint Freeze-Lock & Brittle Gear Shearing [extracted verbatim from the combined masthead line]'

ACTIVE INTERNAL LUBRICANT SUMP CORE (VERBATIM)

* Design Foundation: Rejects local exterior joint-heating jackets. Implements a centralized, hyper-insulated structural Lub-Sump Tank within the primary vehicle chassis. * Pre-Flight Fluid Liquefaction: High-efficiency thermal elements pre-heat heavy-duty industrial greases to a low-viscosity, highly fluid state prior to mechanical rollout. * Conduit Shell Pre-Heating: Phase 61 thin-film carbon sleeves snap-activate along external conduit hulls, raising transmission line temperatures above the -40°C embrittlement threshold to permanently neutralize localized fluid flash-freezing blockages.

THE TWIN-SCREW MECHANICAL PUSH-PULL PUMP (VERBATIM)

* Rationale: Overrides hydraulic fluid pump blowouts caused by extreme sub-zero temperature sludge-viscosity. * Archimedes Drive Infrastructure: Embeds dual mechanical Archimedes Screw Feeds within the core of the arm lines. * Synchronized Circulation Track: Screw A pushes the heated, fluid grease outbound straight through the internal actuator gear teeth. Screw B applies automated pulling leverage to return the cooling fluid back to the central sump in a continuous closed loop.

THE BULBOUS INLINE COMPLIANCE SAFETY GATE (VERBATIM)

* Diagnostic Sensor Pods: Incorporates a widened, bulbous diagnostic terminal along the return fluid conduit. * The Pressure-Movement Audit: Houses automated inline pressure sensors and micro-kinetic displacement testers to measure real-time fluid flow and viscosity baselines. * Motor Ignition Interlock: Prevents primary joint movement until the loop registers smooth fluid circulation above the -40°C safety baseline, ensuring all gears are coated in a warm lubricant cushion.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE FLUID POWER, AFFIRMED AS ORDINARY: $P = F/A$ and $P_{\text{hydraulic}} = \Delta P \cdot Q$ — the vet's own ledger lines. Hydraulic actuation and Archimedes screws are centuries-proven machine families; the phase's engineering is in the thermal management and the circulation topology, not in exotic physics.

THE SUMP DOCTRINE, AFFIRMED: every combustion engine on Earth already solves this exact problem

— oil sumps recirculate warmed lubricant through a cold block. Centralizing the grease heat in one insulated sump and circulating it is the internal-combustion answer applied to actuator lines; local heating jackets treat symptoms at every joint, the sump treats the cause once.

THE TWIN-SCREW, AFFIRMED AS THE RIGHT PUMP FOR SLUDGE: positive-displacement screws move viscous fluid regardless of the pressure-instability that stalls centrifugal and gear pumps on cold grease — push-pull twin screws give the loop a driven return, not a hopeful one.

THE INTERLOCK, AFFIRMED AS DISCIPLINE: pressure and displacement sensors gating motor ignition convert 'warm it up first' from procedure into physics — the arm cannot be driven cold because the machine will not let it. The owed numbers: available pressure, flow, seal performance, temperature envelope.

§3 — THE SYSTEM IN THE LINE

- THE SUMP: a centralized hyper-insulated Lub-Sump Tank in the primary chassis; thermal elements pre-heat the grease to working viscosity.
- THE SCREWS: dual Archimedes screw feeds in the arm-line cores — Screw A pushes heated grease out, Screw B pulls the return.
- THE GATE: the bulbous inline diagnostic terminal — pressure sensors and micro-kinetic displacement testers measuring flow and viscosity in real time.
- THE INTERLOCK: motor ignition locked to the pressure-movement audit — no cold drives.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE SUMP DOCTRINE (seated with the core): treat the cause once, centrally — the engine block already knew how; local jackets treat symptoms forever.
- THE POSITIVE-DISPLACEMENT DOCTRINE (seated with the screws): sludge does not care about impeller curves — a screw moves what a pump cannot.
- THE MACHINE-SAYS-NO DOCTRINE (seated with the interlock): the warm-up is not a procedure to remember; it is a gate the machine enforces.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Build the sump: the insulated Lub-Sump Tank with its thermal elements; heat-up curves

from deep-cold to working viscosity logged.

- STEP 2 — Drive the screws: the twin Archimedes feeds on a test arm line; push-pull flow rates across the temperature envelope measured.
- STEP 3 — Fit the gate: the bulbous diagnostic terminal instrumented; baseline pressure/viscosity maps published.
- STEP 4 — Wire the interlock: ignition gated to the audit; cold-drive attempts logged as denied, not as failures.
- STEP 5 — Freeze the rig: the full arm assembly cycled at the sub-zero design floor; seal performance and the blowout margin measured.
- STEP 6 — Publish the loop: temperatures, flows, pressures, seal figures — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 61 — the thermal-sleeve conduits (GP-IM-061): the sibling thermal-baseline family — heat before flow.
- PHASE 70 — the parallel synchronicity conduits (GP-IM-070): the loop architecture the return line parallels (and the Phase 110 packaging).
- PHASE 41 — the Mule Tugs (GP-IM-041): the arm-line vehicles the sumps serve.
- PHASE 88 — the Nitropril stabilizers (chapter owed): the material feed the screws also serve.

§8 — DO-NOT-BUILD

- NEVER drive a cold arm — the interlock is law; a denied start is the machine working.
- NO local-jacket patches — the sump centralizes the heat; symptom-treating every joint is the failure this phase deletes.
- NEVER trust a pressure-only pump on cold grease — sludge stalls impellers; the screw is the circulation.
- NO unsealed envelopes — seal performance at the temperature floor is a benched number before the arm flies.

§9 — OWED

THE NUMBERS, OWED (the vet's variables): available pressure and flow across the temperature envelope; seal performance at the sub-zero floor; sump heat-up curves; the twin-screw throughput

versus viscosity.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-087, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, tenth of the new 'next 10' shift ('10 more before go to the old lady for dinner'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562 and 564): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 87